

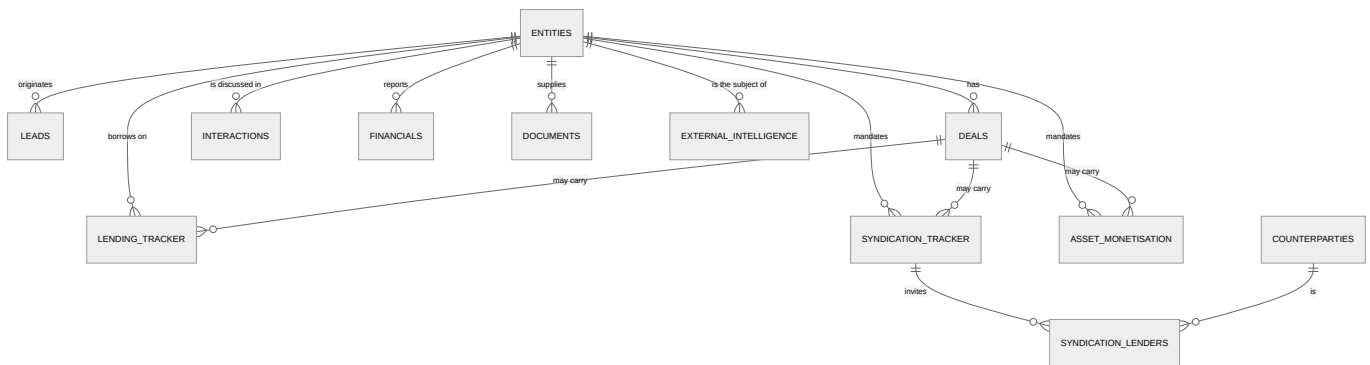
15 — Data Model & ERD

Audience: anyone writing a query, a migration, or a report. **Companion docs:** [08 Register](#) · [04 Running flows](#) · [09 Backup](#) **Code:** `services/register/app/models/` · `docs/SCHEMA.md` for the column-level detail

1. The one commitment: entity-centric

"A single legal entity, CIN-anchored where possible. The spine everything links to: one company appears once and every deal/financial/touchpoint links back to it. This is the entity-centric (not deal-centric) commitment at the heart of the Register." — `models/registry.py`

Deal-centric systems duplicate the company for every transaction, and then nobody can answer "what is our total exposure to this promoter group?" without a reconciliation exercise. Here, `entities` is the spine.



Note the word **"may"** on the deal → tracker edges. `deal_id` is **nullable**; `entity_id` is **not**. A mandate always belongs to a company; it does not always belong to a deal record. That is deliberate, and it is why imported mandates with no Deals-sheet row are valid data rather than orphans.

2. Every table, by domain

~43 tables across 20 model modules.

Master data

Table	Contents
<code>entities</code>	The spine. Companies, promoters, directors, related parties
<code>people</code>	The internal directory — RMs, analysts, heads
<code>counterparties</code>	Banks, NBFCs, DFIs, funds, investors, advisors
<code>ref_values</code>	Every dropdown vocabulary, served from <code>/v1/ref</code>
<code>tenants</code> , <code>tenant_settings</code>	Tenant registry and per-tenant configuration

Origination

Table	Contents
<code>leads</code>	Opportunities before conversion
<code>deals</code>	The commercial funnel record, with the three product flags

The three product books

Table	Contents
<code>lending_tracker</code>	The credit pipeline line
<code>syndication_tracker</code>	One mandate per company
<code>syndication_lenders</code>	One row per bank per mandate — the lender matrix
<code>asset_monetisation</code>	One row per mandate (a company may have several)

Company file

Table	Contents
<code>interactions</code>	The timeline. Typed <i>and</i> VocX-dictated
<code>financials</code>	Statements: audited, provisional, projections, GST, CMA
<code>contracts_assets</code>	PPA, EPC, O&M, offtake, security charge
<code>documents</code> , <code>document_checklist</code>	Files and the required-set checklist
<code>external_intelligence</code>	PULSE items and other intel, RED/AMBER/GREEN
<code>monitoring_reporting</code>	Covenant compliance, security creation, periodic submissions

Credit governance

Table	Contents
<code>sanction_terms</code>	The sanctioned terms
<code>cam_reports</code> , <code>cam_turns</code>	The CAM workbench: the memo and its drafting turns
<code>cp_cs_checklists</code>	The authoritative CP/CS checklist (maker-checker)
<code>governance_evidence</code> , <code>governance_evidence_status</code>	Evidence gating a stage
<code>workflow_decisions</code> , <code>workflow_decision_outbox</code>	The single-winner decision record
<code>change_requests</code>	Stage-change requests awaiting approval

Servicing (LMS) — present, but `LMS_ENABLED=false` today

Table	Contents
<code>loan_accounts</code>	The booked account
<code>loan_ledger_entries</code>	Ledger postings
<code>loan_account_conditions</code>	Post-disbursement conditions
<code>disbursement_tranches</code>	Tranche booking
<code>covenants</code>	Covenant definitions and observations
<code>ews_cases</code>	Early-warning cases

Handover

Table	Contents
<code>advaya_handover_packages</code>	The immutable package, sha256 recorded
<code>advaya_handoffs</code>	The boundary record and acknowledgement

Platform plumbing

Table	Contents
<code>line_assignments</code>	Who owns which row — the basis of every SCOPED decision
<code>import_reconciliation_items</code>	The import quarantine queue
<code>notifications</code> , <code>notification_deliveries</code>	Outbox and delivery attempts
<code>calendar_events</code>	Scheduled follow-ups and meetings
<code>idempotency_keys</code>	Replay protection
<code>number_series</code>	Tracker numbering
<code>audit_log</code>	Append-only. Never updated, never soft-deleted

3. What every business row carries

Every table inheriting `RecordBase` gets the same spine:

```
class RecordBase(Base, TimestampMixin, SoftDeleteMixin):
    """Concrete base for every tenant-aware, versioned, auditable business table."""
    id: Mapped[uuid.UUID] = mapped_column(primary_key=True, default=uuid.uuid4,
                                          server_default=func.gen_random_uuid())
    tenant_id: Mapped[uuid.UUID] = mapped_column(nullable=False, index=True)
    version: Mapped[int] = mapped_column(nullable=False, default=1, server_default="1")
    __mapper_args__ = {"version_id_col": version}
```

Column	Purpose
<code>id</code>	UUID, generated in the database (<code>gen_random_uuid()</code>) as well as in Python
<code>tenant_id</code>	Never null. Indexed. The RLS policy key
<code>version</code>	Optimistic-locking counter — SQLAlchemy bumps and checks it on every UPDATE
<code>created_at</code> , <code>updated_at</code>	<code>TimestampMixin</code>
<code>deleted_at</code>	<code>SoftDeleteMixin</code> — non-null means deleted

So: **every business row is tenant-scoped, versioned, timestamped and soft-deletable, by construction.** You do not opt in.

Soft delete and unique constraints

A soft-deleted row must not hold its unique key hostage:

```
Index("entities_tenant_code", "tenant_id", "code", unique=True,
      postgresql_where=text("deleted_at IS NULL"))
```

Partial unique indexes over live rows only. If you add a unique constraint to a soft-deletable table, do the same.

The audit log is different

`audit_log` inherits plain `Base`, not `RecordBase` — *"append-only. Never updated, never soft-deleted."* Columns: `at`, `actor`, `action`, `resource_type`, `resource_id`, `request_id`, `changes` (JSONB). The `request_id` is the same `X-Request-ID` nginx stamps, so an audit entry can be tied back to a specific HTTP request in the logs.

4. Referential integrity — the `onDelete` choices are a policy

```
# lending_tracker, syndication_tracker, asset_monetisation
entity_id = mapped_column(ForeignKey("entities.id", onDelete="RESTRICT"), nullable=False,
index=True)
deal_id   = mapped_column(ForeignKey("deals.id",      onDelete="SET NULL"),
index=True)

# syndication_lenders
syndication_id = mapped_column(ForeignKey("syndication_tracker.id", onDelete="CASCADE"))
counterparty_id = mapped_column(ForeignKey("counterparties.id",      onDelete="SET NULL"))
```

Each choice says something:

Rule	Meaning
<code>entity_id ... RESTRICT</code> , <code>nullable=False</code>	A company with a book cannot be deleted. Every mandate belongs to a company, always
<code>deal_id ... SET NULL</code>	Deleting a deal record leaves the mandate standing. The mandate is the real thing; the deal record is an origination wrapper
lenders <code>CASCADE</code> from the mandate	A lender row has no meaning without its mandate
lenders <code>SET NULL</code> to counterparty	The bank leaving the master must not erase the history of it being invited

5. The three lifecycle fields

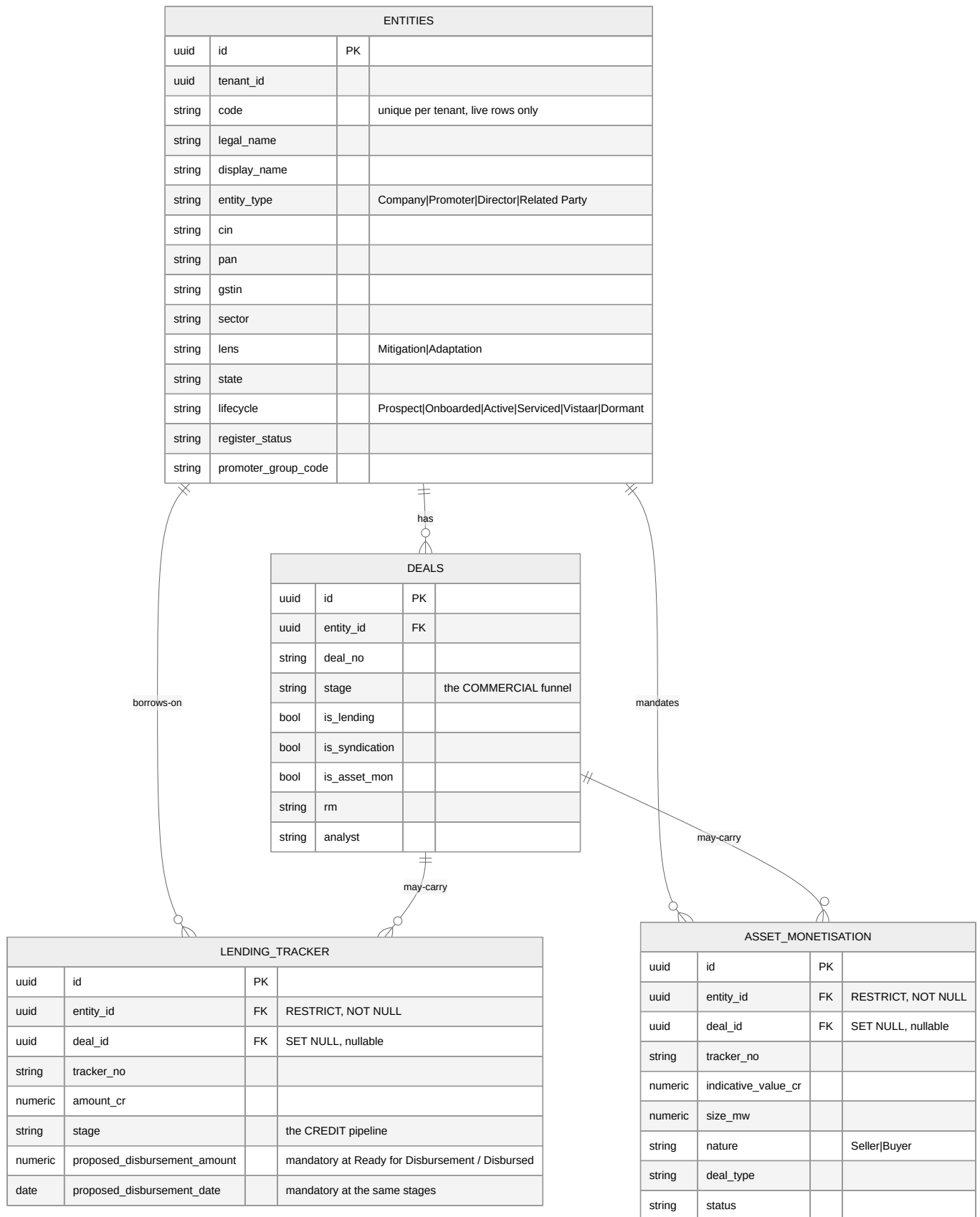
Only five subjects carry an enforced lifecycle. They key off different columns, which is a frequent source of confusion:

Subject	Table	Field
Lead	<code>leads</code>	<code>status</code>
Deal	<code>deals</code>	<code>stage</code>
Lending	<code>lending_tracker</code>	<code>stage</code>
Syndication	<code>syndication_tracker</code>	<code>status</code>
AssetMonetisation	<code>asset_monetisation</code>	<code>status</code>

Lending uses `stage`; syndication and AM use `status`. `policy.py::_STAGE_FIELD` is the map, and a key that does not match the API subject string **silently disables every lifecycle rule for that resource**.

Deals also carry `credit_stage_legacy` — historical values parked there when the deal-level credit stage was deprecated in favour of the lending line's. **Do not use it for new work.**

6. Entity relationships in detail



Why grids join client-side

A tracker row carries `entity_id` and `deal_id` — **not** a company name. The API keeps rows normalised, and `services/atlas/ui/src/services/nameResolver.ts` fetches two small lookup maps (entity →

name+code, deal → number+entity), caches them 60 s and joins in the browser.

It **fails soft**: a resolver outage leaves the joined columns blank rather than failing the grid. And it falls back to the row's own `entity_id` when the deal lookup yields nothing — which is what makes deal-less imported mandates display correctly.

7. Governance and decisions

WORKFLOW_DECISIONS			
uuid	id	PK	
string	workflow_id		
string	kind		approve reject committee syndication am
string	decided_by		the FIRST decider wins
string	decided_by_id		
jsonb	roles		
jsonb	operations		
text	note		
timestampz	at		

GOVERNANCE_EVIDENCE		
uuid	id	PK
string	subject_type	
uuid	subject_id	
string	evidence_kind	
string	reference	
string	sha256	

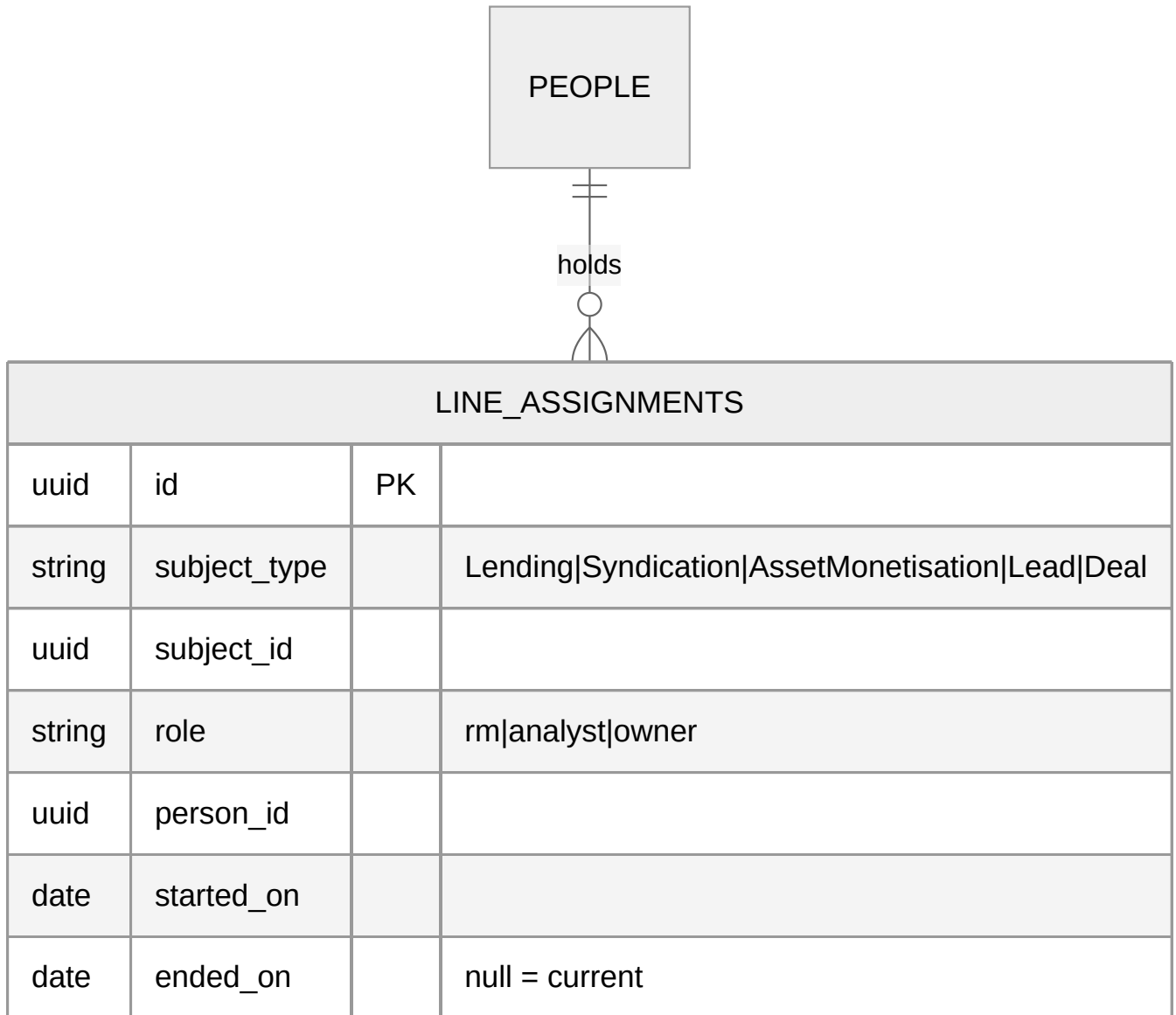
CHANGE_REQUESTS		
uuid	id	PK
string	subject_type	
uuid	subject_id	
string	target_stage	
string	requested_by	
string	status	

ADVAYA_HANDOVER_PACKAGES			
uuid	id	PK	
uuid	lending_id	FK	
jsonb	package		IMMUTABLE
string	payload_sha256		
string	status		

`workflow_decisions` is the **single-winner** record and the sole authority for a workflow decision. A Temporal signal is only a nudge; the worker reads identity, note and outcome from this row. Two approvers submitting the same outcome produce one record naming the first, and the run and the database can never disagree.

`governance_evidence` carries a `sha256` because "the committee approved this" must name *which document* — an evidence reference without a hash can be swapped after the fact.

8. Scoping and assignment



This one table is the basis of every `SCOPED` authorization decision. When the gateway says `X-Authz-Decision: SCOPED`, the Register asks: *is this row assigned to this person, or in their vertical?* Assignments are **time-bounded** (`ended_on`), so history is preserved when ownership moves — you can answer "who owned this in March?".

9. Tenant isolation in the schema

Every business table has `tenant_id NOT NULL, index`. With `REGISTER_ENFORCE_RLS=true`, `app/db/apply_qls.py` applies and force-converges a row-level-security policy per table at startup.

```
-- the shape (see apply_qls.py for the real thing)
ALTER TABLE entities ENABLE ROW LEVEL SECURITY;
ALTER TABLE entities FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON entities
  USING (tenant_id = current_setting('prism.tenant_id')::uuid);
```


Fail-closed: if the policy cannot be established, the service refuses to serve rather than serving everything. A forgotten `WHERE tenant_id = ...` in application code becomes a non-event instead of a cross-tenant leak.

Adding a table means adding its RLS policy. A new tenant-scoped table without one is a hole, and it will not announce itself.

10. Indexing conventions

Pattern	Example
Every FK is indexed	<code>entity_id</code> , <code>deal_id</code> , <code>syndication_id</code>
<code>tenant_id</code> is always indexed	<code>RecordBase.tenant_id</code>
Composite <code>(tenant_id, <filter>)</code> for common filters	<code>ix_entities_tenant_sector</code> , <code>ix_entities_tenant_cin</code>
Partial unique over live rows	<code>entities_tenant_code ... WHERE deleted_at IS NULL</code>
Naming convention is centralised	<code>NAMING_CONVENTION</code> in <code>evam_backend_core/db/base.py</code>

Filters exposed on the API are whitelisted per resource and should have a matching composite index. An unindexed filterable column is a latent full scan on a growing table.

11. Migrations

Alembic, per service:

```
services/register/migrations/versions/
services/access/migrations/versions/
```

Applied automatically as services start (`alembic upgrade head`).

`prism-deploy.sh` diffs both directories between the live and new trees and **announces the delta before the upgrade builds anything** — see [10 §4](#).

Writing a migration that can be rolled back

Change	Rollback-safe?	Guidance
Add a table	Yes	Old code ignores it
Add a nullable column	Yes	
Add NOT NULL with a default	Yes	
Add NOT NULL without a default	No	Old code's INSERTs fail
Rename a column	No	Two steps: add new + backfill + dual-write, then drop in a later release
Drop a column or table	No	Deprecate first, drop a release later
Change a type	Depends	Widening is usually fine; narrowing is not

Expand/contract is the rule. A release should be rollback-safe on its own; the destructive half comes one release later, once you are sure.

12. Querying the book directly

```
DC="docker compose -f deploy/compose/docker-compose.yml"
$DC exec postgres psql -U prism -d register
```

```
-- the book at a glance
select 'entities' t, count(*) from entities where deleted_at is null
union all select 'deals', count(*) from deals where deleted_at is null
union all select 'lending', count(*) from lending_tracker where deleted_at is null
union all select 'syndication', count(*) from syndication_tracker where deleted_at is null
union all select 'asset_mon', count(*) from asset_monetisation where deleted_at is null
union all select 'interactions', count(*) from interactions where deleted_at is null;

-- lending pipeline by stage
select stage, count(*), round(sum(amount_cr)::numeric, 2) as cr
from lending_tracker where deleted_at is null group by stage order by 2 desc;

-- mandates with no deal record (legal – usually imported)
select 'lending' src, count(*) from lending_tracker where deal_id is null and deleted_at is null
union all select 'syndication', count(*) from syndication_tracker where deal_id is null and deleted_at is null
union all select 'asset_mon', count(*) from asset_monetisation where deal_id is null and deleted_at is null;

-- the import quarantine
select subject_type, status, count(*) from import_reconciliation_items group by 1, 2;

-- who owns what, currently
select p.name, la.subject_type, count(*)
from line_assignments la join people p on p.id = la.person_id
where la.ended_on is null group by 1, 2 order by 3 desc;
```

Two habits worth keeping: **always filter** `deleted_at is null` (soft delete means rows stay), and **always scope by** `tenant_id` in a multi-tenant deployment — a `psql` session is not going through the application's tenant scoping.